

Case 1

A big company has decided to store information about its projects and employees in a database. The company has wisely chosen to hire you as a database designer. Prepare an E-R diagram for this Company according to The following Description:

- ◆ The company has a number of **employees** each employee has **SSN, Birth Date, Gender and Name** which represented as **Fname and Lname**.
- ◆ The company has a set of **departments** each department has a set of attributes **.0DName, DNUM (unique) and locations**.
- ◆ Employees work in several **projects** each project has **Pname, PNumber as an identifier, Location and City**.
- ◆ Each employee may have a set of **dependent**; each dependent has **Dependent Name (unique), Gender, and Birth Date**.
Note: if the employee left the company no needs to store his dependents info
- ◆ For each Department, there is always one employee assigned **to manage** that Department and each manager has a **hiring Date**
- ◆ Department may have employees but employee must work on Only One department
- ◆ Each department may have a set of projects and each project must assigned to one department
- ◆ Employees work in several projects and each project has several employees and each employee has a number of working hours in each project
- ◆ Each employee has a supervisor

Case 2

ITI has decided to store information about its Students and curriculums in a database. ITI has wisely chosen to hire you as a database designer. Prepare an E-R diagram for ITI that describes its activities according to The following Description:

- ✦ ITI has a number of students in different departments(tracks), each student has St_id(unique),studentName(Fname,Lname),St_age,st_address
- ✦ Each department has dept_id(unique), dep_name
- ✦ Students takes many Courses, each course has crs_id, crs_name, crs_duration and crs_Description
- ✦ Student has a grade in each course
- ✦ Each department contains a set of instructor and each instructor is located in one department, each instructor has ins_id, ins_name, ins_salary, ins_hourRate, ins_bouns,ins_address
- ✦ For each Department, there is always one instructor assigned to manage that Department and each manager has a hiring Date
- ✦ Instructor may teach many courses and each course may be conducted by many instructors, each instructor has evaluation in each course
- ✦ Course is classified under one topic, each topic may have many courses, and each topic has top_name and top_id